Sequences — problem 9.26

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A sequence a_n is defined by $a_1 = 1$, $a_2 = 12$, $a_3 = 20$, $a_{n+3} = 2a_{n+2} + 2a_{n+1} - a_n$, $n \geq 0$. Prove that, for every n, the integer $1 + 4a_n a_{n+1}$ is a square.